Meta-Learning for Accelerated PDE Discovery

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Abstract—These instructions give you guidelines for preparing your project report. DO NOT change any settings, such as margins and font sizes. Just use this as a template and modify the contents into your final paper. An abstract should be 100 to 200 words, and should clearly state the nature and significance of the paper. Abstracts must not include mathematical expressions or bibliographic references. Please note that abstracts are formatted as left justified in our editing template (as shown here).

Index Terms—Keywords should be taken from the taxonomy (http://www.computer.org/mc/keywords/keywords.htm). Keywords should closely reflect the topic and should optimally characterize the paper. Use about four key words or phrases in alphabetical order, separated by commas (there should not be a period at the end of the index terms)

1 Introduction

Partial differential equations are the primary mathematical language for modelling spatiotemporal dynamics in physical systems, from fluid turbulence and chemical pattern formation to heat conduction and biological signalling. While the governing equations for many systems are well established, there exist phenomena whose underlying PDEs are unknown or only partially understood. Data-driven PDE discovery addresses this gap: given observations of a system's evolution, can we recover the governing equation directly from data?

Raissi [1] introduced the Deep Hidden Physics framework, in which a neural network — the PDE operator network — learns the nonlinear mapping from spatial derivatives to the time derivative:

$$u_t = \mathcal{N}(u, u_x, u_y, u_{xx}, u_{yy}, \ldots) \tag{1}$$

Given solution data and its derivatives, the network is trained to approximate the right-hand side of the PDE. Once trained, the learned operator encodes the discovered physics: its weights implicitly represent the PDE coefficients, which can be extracted via Jacobian analysis.

However, this approach requires training a new network from scratch for each system encountered. Every new set of physical parameters — a different viscosity, a different diffusion coefficient — demands a full training run. This is expensive and fails to exploit the structural similarities that exist across related systems. For instance, a network trained on one viscosity of Navier-Stokes learns representations relevant to nearby viscosities, yet none of this knowledge transfers to the next training run.

Model-Agnostic Meta-Learning (MAML) [2] offers a solution to this transfer problem. Rather than learning a single set of weights for one system, MAML learns a meta-parameter *initialisation* θ^* — a starting point in the loss landscape from which the network can rapidly adapt to any system within a task family of one PDE. This solves the expensive training of a large model or scarce dataset.

In this paper MAML will be used to find the meta-parameter *initialisation* θ^* for PDE operator network $\mathcal{N}_\rho$ for the PDE ρ. $\mathcal{N}_\rho$ is initialised with with random θ_0 parameters. $\tau(\rho)$ is the task distribution for the ρ family, and contains tasks of $\{\vec{x}_{(i)}, \vec{y}_{(i)}\}$ pairs. $\vec{x}_{(i)}$ is the input spatial derivatives features, where $\vec{y_{(i)}}$ is the target time derivatives.

2 Related Work

This section presents a survey of existing work on the problems that this project addresses. It should be between 2 to 3 pages in length. The rest of this section shows the formats of subsections as well as some general formatting information for tables, figures, references and equations.

2.1 Main Text

The font used for the main text should be Palatino and the font size should be 9.5. The first line of all paragraphs should be indented by 0.5cm, except for the first paragraph of each section, subsection, subsubsection etc. (the paragraph immediately after the header) where no indentation is needed.

2.2 Figures and Tables

In general, figures and tables should not appear before they are cited. Place figure captions below the figures; place table titles above the tables. If your figure has two parts, for example, include the labels "(a)" and "(b)" as part of the artwork. Please verify that figures and tables you mention in the text actually exist. Make sure that all tables and figures are numbered as shown in Table 1 and Figure 1 below.

2.3 References

The list of cited references should appear at the end of the report, listed by order of appearance in the paper. The required style is to note citations in individual brackets, followed by a comma, e.g. "[1], [5]" (as opposed to the more common "[1, 5]" form.) Citation ranges should be formatted as follows: [1], [2], [3], [4] (as opposed to [1]-[4]). When citing a section in a book, please give the relevant page numbers

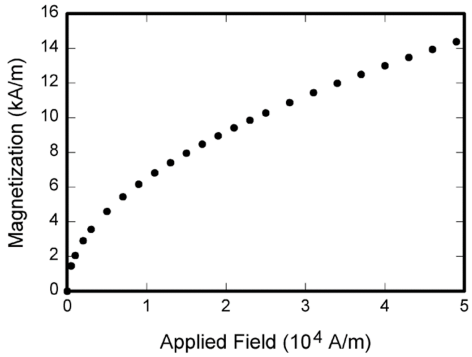

Fig. 1. Magnetization as a function of applied field. There is a period after the figure number, followed by two spaces. It is good practice to explain the significance of the figure in the caption

[2]. In sentences, refer simply to the reference number, as in [3]. Do not use "Ref. [3]" or "reference [3]" At the beginning of a sentence use the author names instead of "Reference [3]," e.g., "Smith and Smith [3] show ... .".

3 Methodology

This section presents the solutions to the problems in detail. The design and implementation details should all be placed in this section. You may create a number of sub-sections, each focusing on one issue.

This section should be between 4 to 6 pages in length.

TABLE 1
Units for Magnetic Properties

Symbol	Quantity	Conversion from Gaussian and CGS EMU to SI [a]
Φ	magnetic flux	1 Mx $\rightarrow 10^{-8}$ Wb $= 10^{-8}$ V·s
B	magnetic flux density, magnetic induction	1 G $\rightarrow 10^{-4}$ T $= 10^{-4}$ Wb/m^2
H	magnetic field strength	1 Oe $\rightarrow 10^3/(4\pi)$ A/m
m	magnetic moment	1 erg/G = 1 emu $\rightarrow 10^{-3}$ A·m^2 $= 10^{-3}$ J/T
M	magnetization	1 erg/(G·cm^3) = 1 emu/cm^3 $\rightarrow 10^3$ A/m
$4\pi M$	magnetization	1 G $\rightarrow 10^3/(4\pi)$ A/m
σ	specific magnetization	1 erg/(G·g) = 1 emu/g $\rightarrow$ 1 A·m^2/kg
j	magnetic dipole moment	1 erg/G = 1 emu $\rightarrow 4\pi \times 10^{-10}$ Wb·m
J	magnetic polarization	1 erg/(G·cm^3) = 1 emu/cm^3 $\rightarrow 4\pi \times 10^{-4}$ T
χ, κ	susceptibility	$1 \rightarrow 4\pi$
χ_ρ	mass susceptibility	1 cm^3/g $\rightarrow 4\pi \times 10^{-3}$ m^3/kg
μ	permeability	$1 \rightarrow 4\pi \times 10^{-7}$ H/m $= 4\pi \times 10^{-7}$ Wb/(A·m)
μ_r	relative permeability	$\mu \rightarrow \mu_r$
w, W	energy density	1 erg/cm^3 $\rightarrow 10^{-1}$ J/m^3
N, D	demagnetizing factor	$1 \rightarrow 1/(4\pi)$

Statements that serve as captions for the entire table do not need footnote letters. E.g. Mx = maxwell, G = gauss, Oe = oersted; Wb = weber, V = volt, s = second, T = tesla, m = meter, A = ampere, J = joule, kg = kilogram, H = henry.

TABLE 2
Summary of Page Lengths for Sections

Section	Number of Pages
I. Introduction	2
II. Related Work	2-3
III. Methodology	4-6
IV. Results	2
V. Evaluation	1-2
VI. Conclusion	1

4 Results

This section presents the results of the solutions. It should include information on experimental settings. The results should demonstrate the claimed benefits/disadvantages of the proposed solutions.

This section should be around 2 pages in length.

5 Evaluation

This section should be between 1 to 2 pages in length.

6 Conclusion

This section summarises the main points of this paper. Do not replicate the abstract as the conclusion. A conclusion might elaborate on the importance of the work or suggest applications and extensions. This section should be no more than 1 page in length.

The page lengths given for each section are indicative and will vary from project to project but should not exceed the upper limit. A summary is shown in Table 2.

REFERENCES

[1] M. Raissi, "Deep Hidden Physics Models: Deep Learning of Nonlinear Partial Differential Equations," *Journal of Machine Learning Research*, vol. 19, no. 25, pp. 1–24, 2018.

[2] C. Finn, P. Abbeel, and S. Levine, "Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks," in *Proc. 34th Int. Conf. Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 70, pp. 1126–1135, 2017.